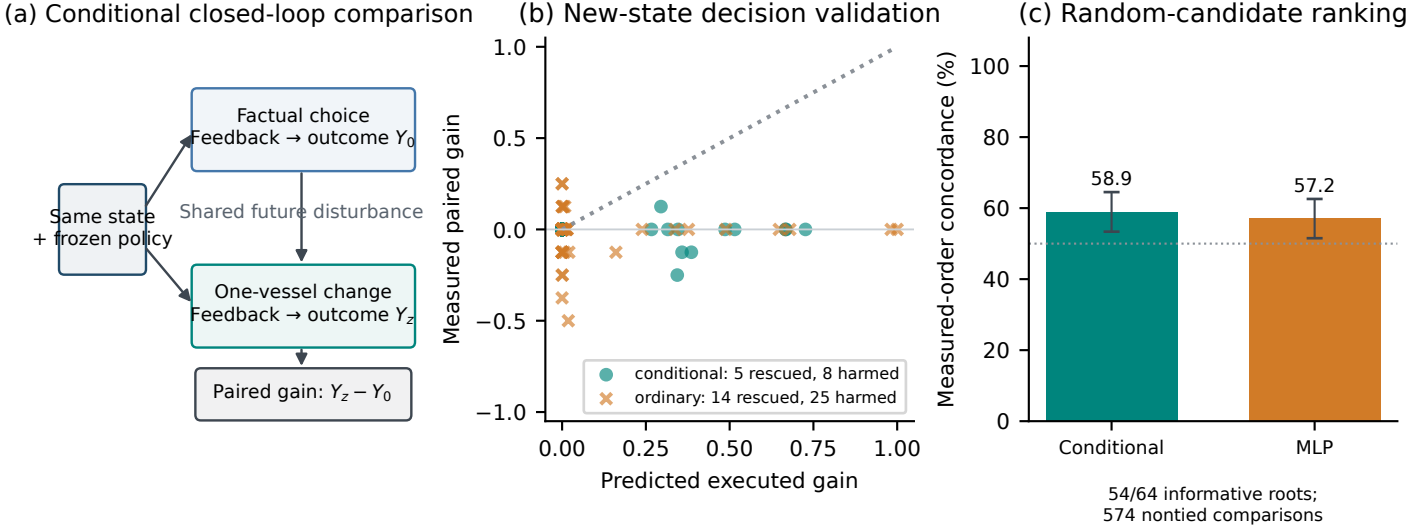

(a) Conditional closed-loop comparison
Factual choice
Feedback → outcome Y_0
Same state
+ frozen policy
Shared future disturbance
One-vessel change
Feedback → outcome Y_z
Paired gain: $Y_z - Y_0$
(b) New-state decision validation
Measured paired gain
1.0
0.5
0.0
−0.5
−1.0
0.00
0.25
0.50
0.75
1.00
Predicted executed gain
conditional: 5 rescued, 8 harmed
ordinary: 14 rescued, 25 harmed
(c) Random-candidate ranking
Measured-order concordance (%)
100
80
60
40
20
0
58.9
57.2
Conditional
MLP
54/64 informative roots;
574 nontied comparisons